2da Lista de Problemas y Ejercicios con Teoría de la Computación

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Autómatas Finitos

En los problemas 94-123, debe hallar un Autómata Finito NO Determinista (AFD) que distinga las palabras del lenguaje que se describe con el alfabeto que se indica. Debe representar la respuesta usando el diagrama de transiciones y la tabla de transiciones. Los alfabetos utilizados son:

$$\Sigma_1 = \{0, 1, 2\}$$

$$\Sigma_2 = \{a, b, c, 0, 1\}$$

$$\Sigma_3 = \{a, b\}$$

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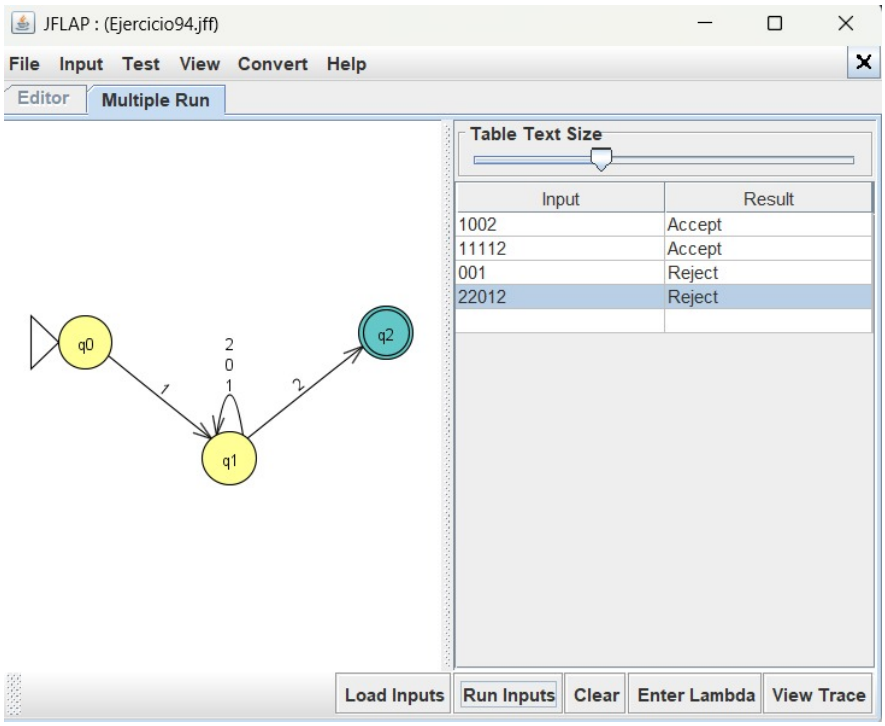
Problemas

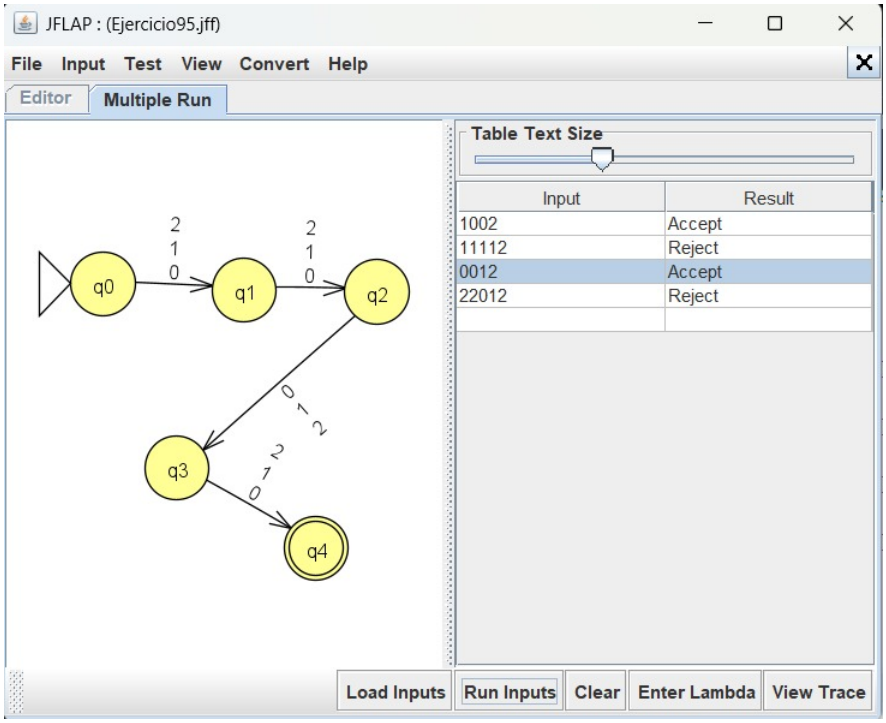
94. Las palabras de Σ_1 que comienzan con 1 y terminan con 2.

Estado	0	1	2
$\rightarrow q_0$	$\emptyset$	$\{q_1\}$	$\emptyset$
q_1	$\{q_1\}$	$\{q_1\}$	$\{q_1, q_2\}$
$*q_2$	$\emptyset$	$\emptyset$	$\emptyset$



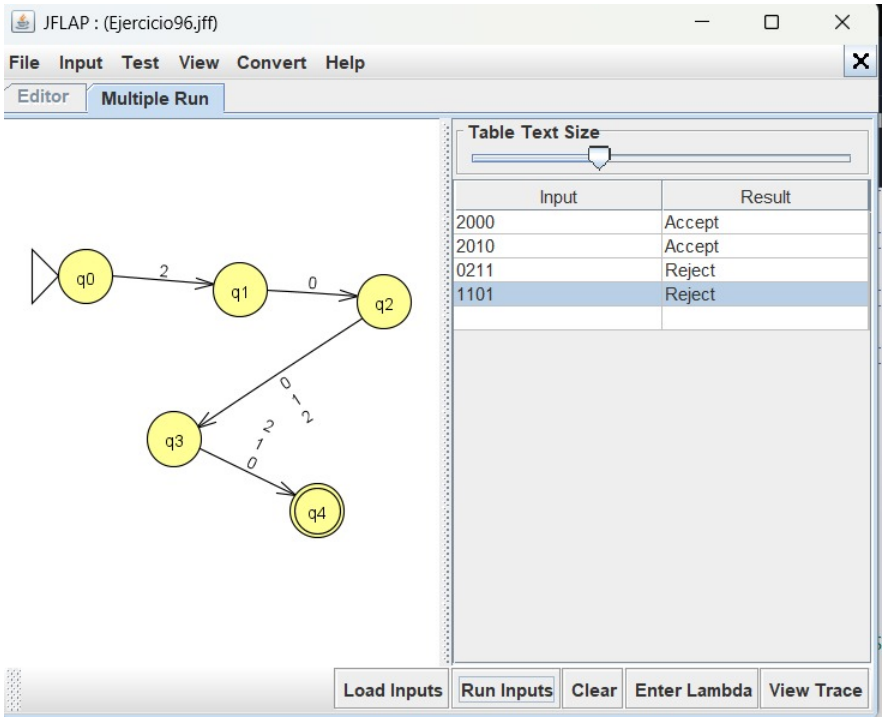
95. Las palabras de Σ_1 que tienen longitud 4.

Estado	0	1	2
$\rightarrow q_0$	$\{q_1\}$	$\{q_1\}$	$\{q_1\}$
q_1	$\{q_2\}$	$\{q_2\}$	$\{q_2\}$
q_2	$\{q_3\}$	$\{q_3\}$	$\{q_3\}$
q_3	$\{q_4\}$	$\{q_4\}$	$\{q_4\}$
$*q_4$	$\emptyset$	$\emptyset$	$\emptyset$



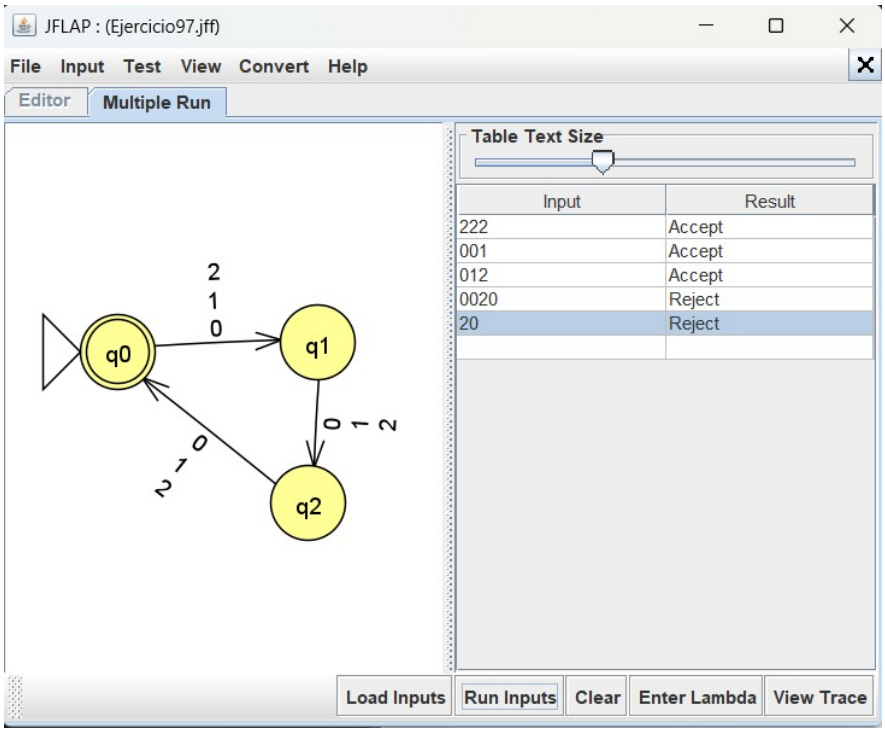
96. Las palabras de Σ_1 con prefijo 20 y longitud 4.

Estado	0	1	2
$\rightarrow q_0$	$\emptyset$	$\emptyset$	$\{q_1\}$
q_1	$\{q_2\}$	$\emptyset$	$\emptyset$
q_2	$\{q_3\}$	$\{q_3\}$	$\{q_3\}$
q_3	$\{q_4\}$	$\{q_4\}$	$\{q_4\}$
$*q_4$	$\emptyset$	$\emptyset$	$\emptyset$



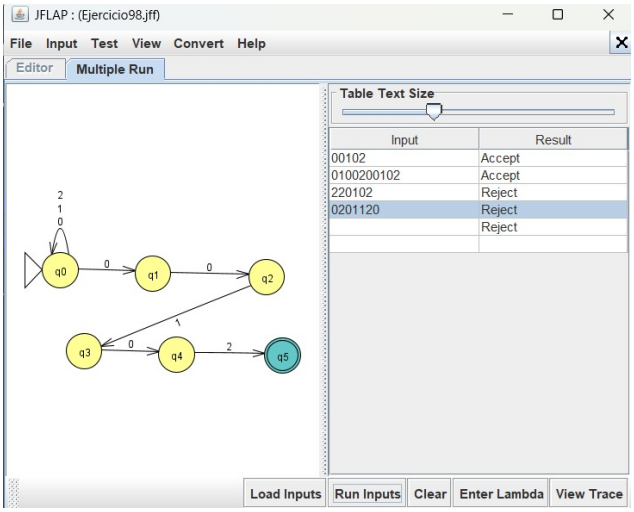
97.* Las palabras de Σ_1 cuya longitud es múltiplo de 3.

Estado	0	1	2
$\rightarrow *q_0$	$\{q_1\}$	$\{q_1\}$	$\{q_1\}$
q_1	$\{q_2\}$	$\{q_2\}$	$\{q_2\}$
q_2	$\{q_0\}$	$\{q_0\}$	$\{q_0\}$



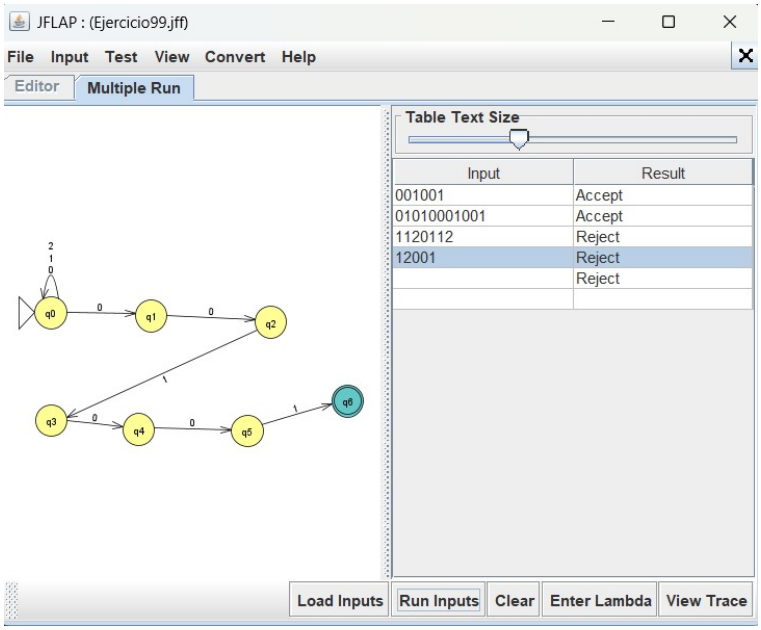
98. Las palabras de Σ_1 que terminan con la cadena 00102.

Estado	0	1	2
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0\}$	$\{q_0\}$
q_1	$\{q_2\}$	$\emptyset$	$\emptyset$
q_2	$\emptyset$	$\{q_3\}$	$\emptyset$
q_3	$\{q_4\}$	$\emptyset$	$\emptyset$
q_4	$\emptyset$	$\emptyset$	$\{q_5\}$
$*q_5$	$\emptyset$	$\emptyset$	$\emptyset$



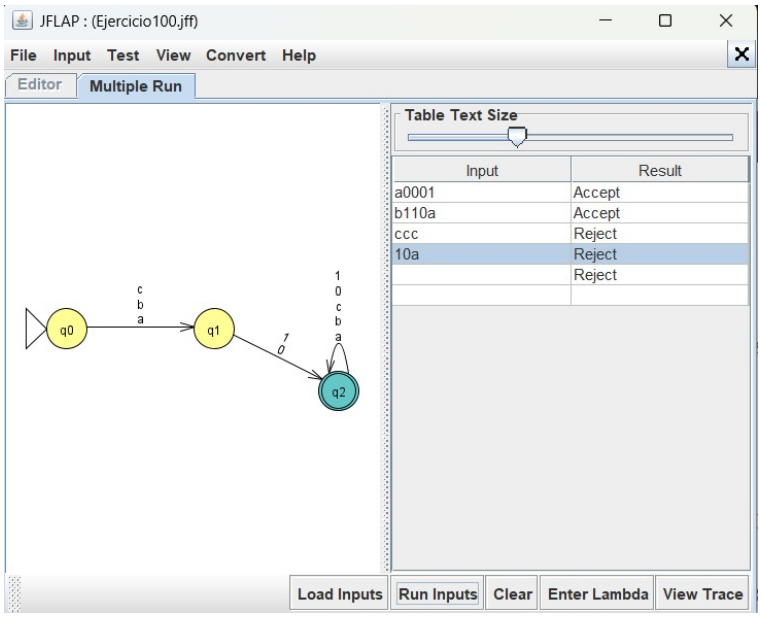
99.* Las palabras de Σ_1 que terminan con la cadena 001001.

Estado	0	1	2
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0\}$	$\{q_0\}$
q_1	$\{q_2\}$	$\emptyset$	$\emptyset$
q_2	$\emptyset$	$\{q_3\}$	$\emptyset$
q_3	$\{q_4\}$	$\emptyset$	$\emptyset$
q_4	$\{q_5\}$	$\emptyset$	$\emptyset$
q_5	$\emptyset$	$\{q_6\}$	$\emptyset$
$*q_6$	$\emptyset$	$\emptyset$	$\emptyset$



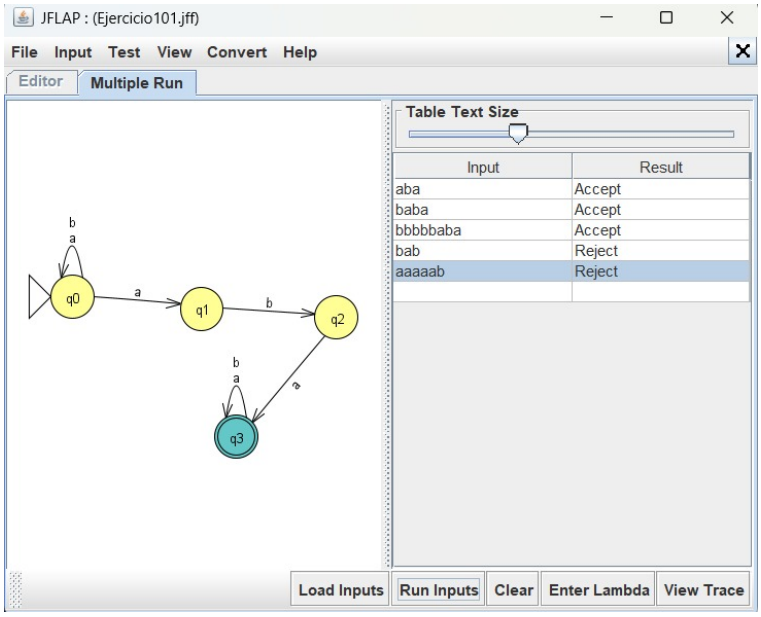
100. Las palabras de Σ_2 donde el primer símbolo es una letra y el segundo es un dígito.

Estado	a	b	c	0	1
$\rightarrow q_0$	$\{q_1\}$	$\{q_1\}$	$\{q_1\}$	$\emptyset$	$\emptyset$
q_1	$\emptyset$	$\emptyset$	$\emptyset$	$\{q_2\}$	$\{q_2\}$
$*q_2$	$\{q_2\}$	$\{q_2\}$	$\{q_2\}$	$\{q_2\}$	$\{q_2\}$



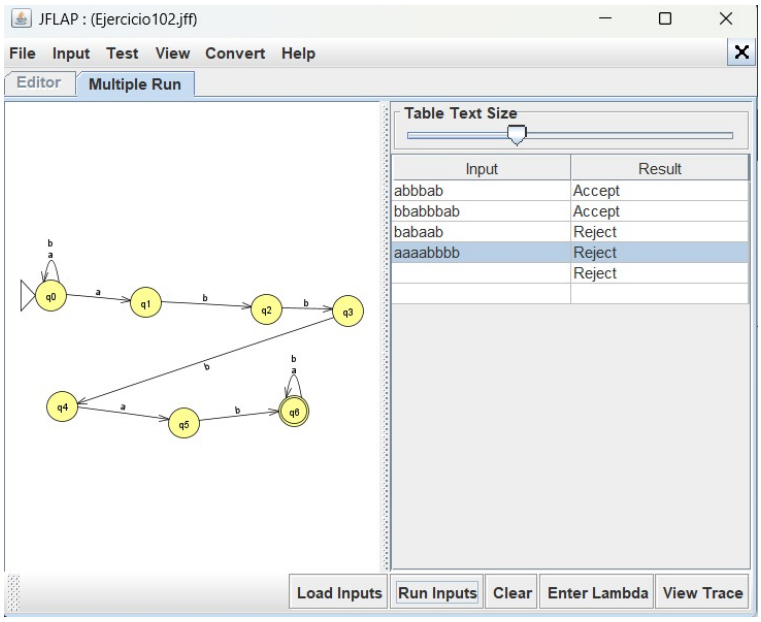
101. Las palabras de Σ_3 que contienen la cadena aba.

Estado	a	b
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0\}$
q_1	$\emptyset$	$\{q_2\}$
q_2	$\{q_3\}$	$\emptyset$
$*q_3$	$\{q_3\}$	$\{q_3\}$



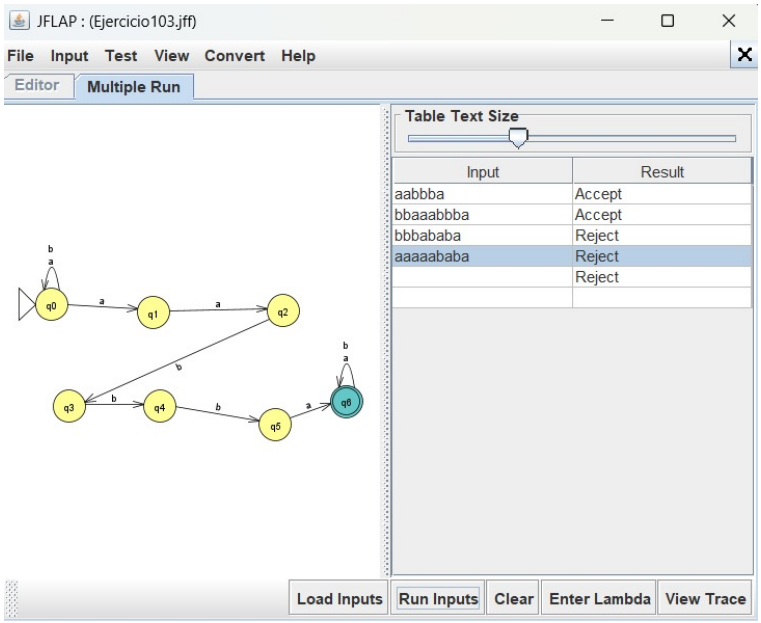
102.* Las palabras de Σ_3 que contienen la cadena abbbab.

Estado	a	b
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0\}$
q_1	$\emptyset$	$\{q_2\}$
q_2	$\emptyset$	$\{q_3\}$
q_3	$\emptyset$	$\{q_4\}$
q_4	$\{q_5\}$	$\emptyset$
q_5	$\emptyset$	$\{q_6\}$
$*q_6$	$\{q_6\}$	$\{q_6\}$



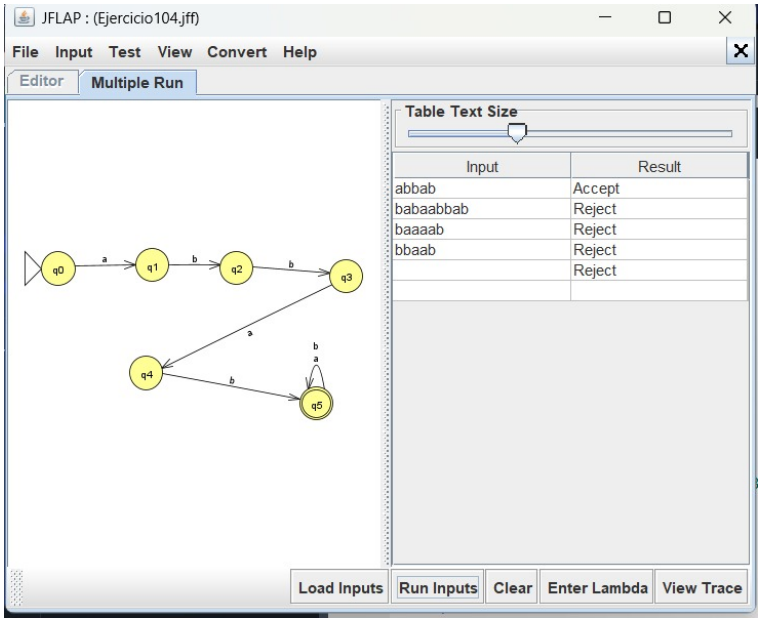
103.* Las palabras de Σ_3 que contienen la cadena aabbba.

Estado	a	b
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0\}$
q_1	$\{q_2\}$	$\emptyset$
q_2	$\emptyset$	$\{q_3\}$
q_3	$\emptyset$	$\{q_4\}$
q_4	$\emptyset$	$\{q_5\}$
q_5	$\{q_6\}$	$\emptyset$
$*q_6$	$\{q_6\}$	$\{q_6\}$



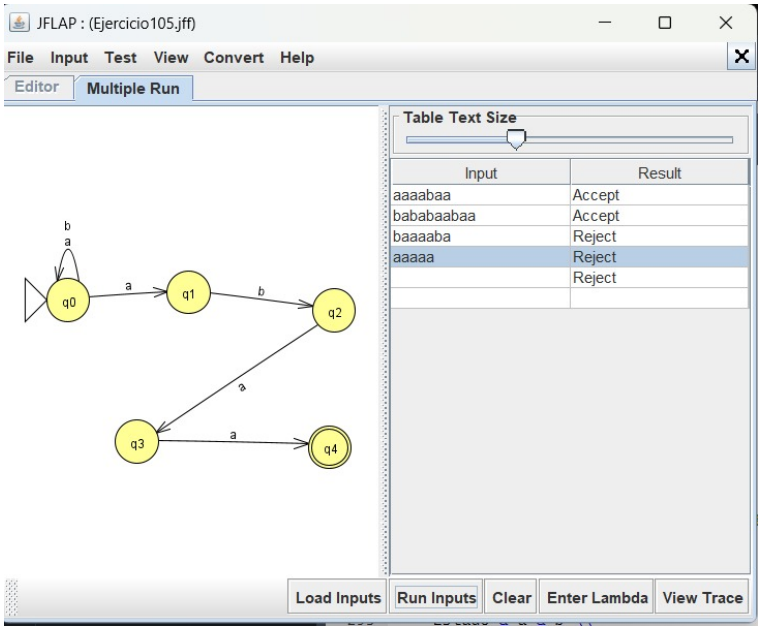
104. Las palabras de Σ_3 que inician con la cadena abbab.

Estado	a	b
$\rightarrow q_0$	$\{q_1\}$	$\emptyset$
q_1	$\emptyset$	$\{q_2\}$
q_2	$\emptyset$	$\{q_3\}$
q_3	$\{q_4\}$	$\emptyset$
q_4	$\emptyset$	$\{q_5\}$
$*q_5$	$\{q_5\}$	$\{q_5\}$



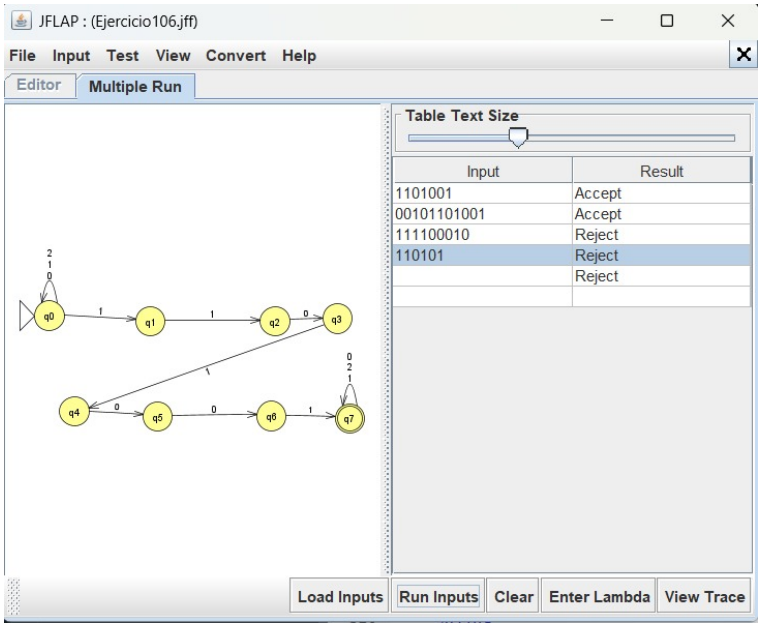
105. Las palabras de Σ_3 que terminan con la cadena abaa.

Estado	a	b
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0\}$
q_1	$\emptyset$	$\{q_2\}$
q_2	$\{q_3\}$	$\emptyset$
q_3	$\{q_4\}$	$\emptyset$
$*q_4$	$\emptyset$	$\emptyset$



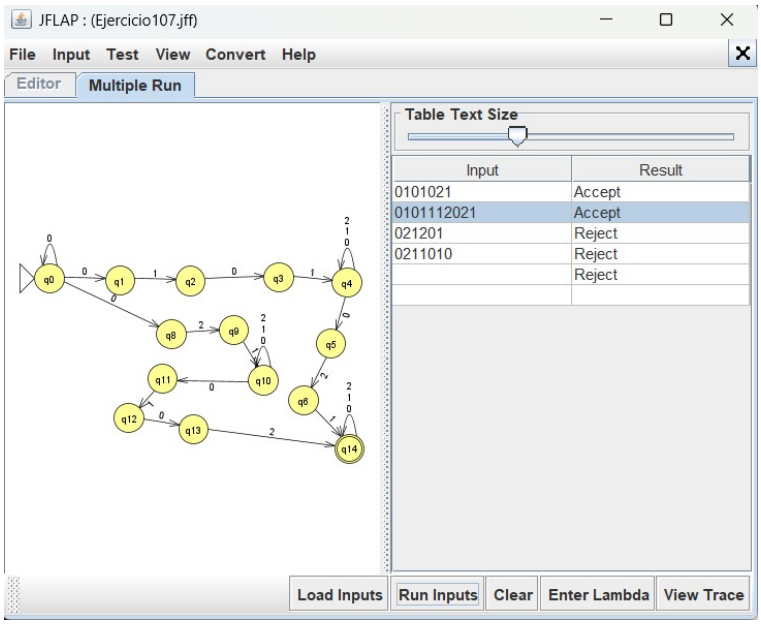
106.* Las palabras de Σ_1 que incluyen la subcadena 1101001.

Estado	0	1	2
$\rightarrow q_0$	$\{q_0\}$	$\{q_0, q_1\}$	$\{q_0\}$
q_1	$\emptyset$	$\{q_2\}$	$\emptyset$
q_2	$\{q_3\}$	$\emptyset$	$\emptyset$
q_3	$\emptyset$	$\{q_4\}$	$\emptyset$
q_4	$\{q_5\}$	$\emptyset$	$\emptyset$
q_5	$\{q_6\}$	$\emptyset$	$\emptyset$
q_6	$\emptyset$	$\{q_7\}$	$\emptyset$
$*q_7$	$\{q_7\}$	$\{q_7\}$	$\{q_7\}$



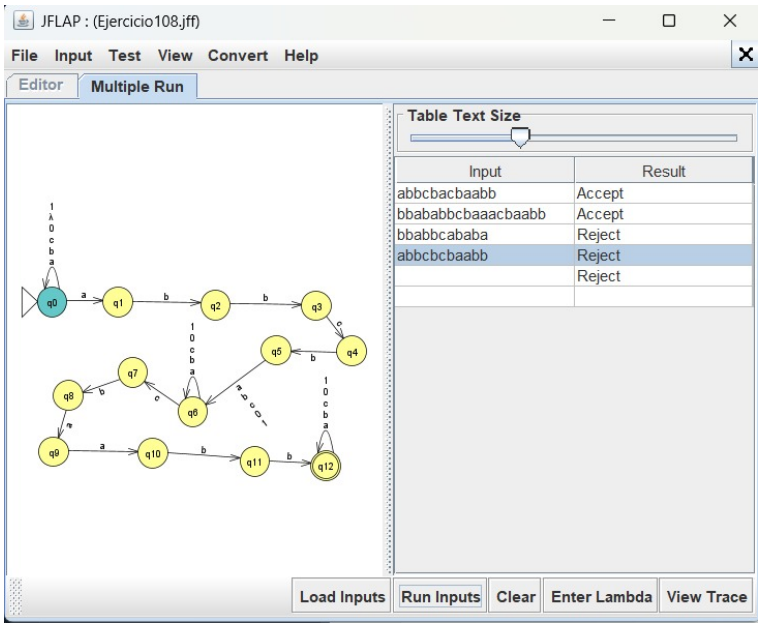
107.* Las palabras de Σ_1 que incluyen al menos una vez 0101 y al menos una vez 021.

Estado	0	1	2
$\rightarrow q_0$	$\{q_0, q_1, q_8\}$	$\{q_0\}$	$\{q_0\}$
q_1	$\emptyset$	$\{q_2\}$	$\emptyset$
q_2	$\{q_3\}$	$\emptyset$	$\emptyset$
q_3	$\emptyset$	$\{q_4\}$	$\emptyset$
q_4	$\{q_4, q_5\}$	$\{q_4\}$	$\{q_4\}$
q_5	$\emptyset$	$\emptyset$	$\{q_6\}$
q_6	$\emptyset$	$\{q_{14}\}$	$\emptyset$
q_8	$\emptyset$	$\emptyset$	$\{q_9\}$
q_9	$\emptyset$	$\{q_{10}\}$	$\emptyset$
q_{10}	$\{q_{10}, q_{11}\}$	$\{q_{10}\}$	$\{q_{10}\}$
q_{11}	$\emptyset$	$\{q_{12}\}$	$\emptyset$
q_{12}	$\{q_{13}\}$	$\emptyset$	$\emptyset$
q_{13}	$\emptyset$	$\{q_{14}\}$	$\emptyset$
$*q_{14}$	$\{q_{14}\}$	$\{q_{14}\}$	$\{q_{14}\}$



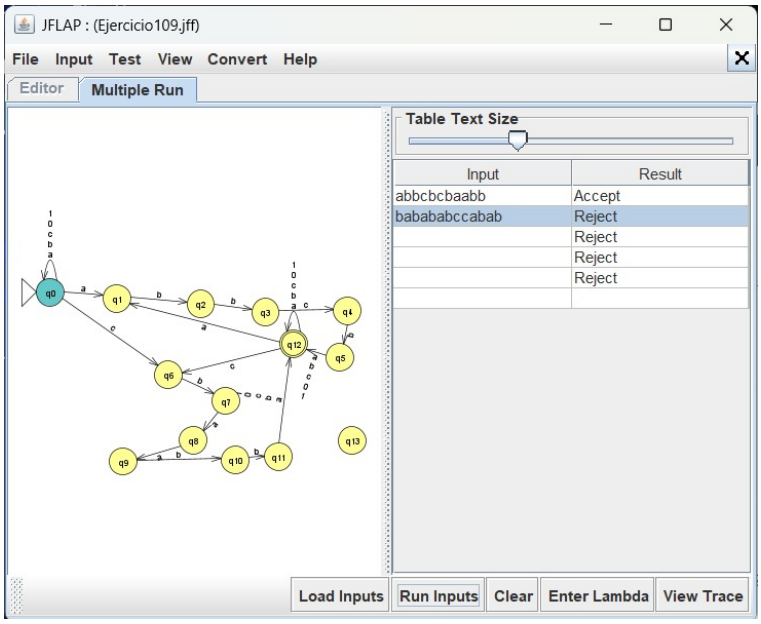
108. Las palabras de Σ_2 que incluyen al menos una vez abbcb y al menos una vez cbaabb, sin traslape.

Estado	a	b	c	0	1
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0\}$	$\{q_0\}$	$\{q_0\}$	$\{q_0\}$
q_1	$\emptyset$	$\{q_2\}$	$\emptyset$	$\emptyset$	$\emptyset$
q_2	$\emptyset$	$\{q_3\}$	$\emptyset$	$\emptyset$	$\emptyset$
q_3	$\emptyset$	$\emptyset$	$\{q_4\}$	$\emptyset$	$\emptyset$
q_4	$\emptyset$	$\{q_5\}$	$\emptyset$	$\emptyset$	$\emptyset$
q_5	$\{q_6\}$	$\{q_6\}$	$\{q_6\}$	$\{q_6\}$	$\{q_6\}$
q_6	$\{q_6\}$	$\{q_6\}$	$\{q_6, q_7\}$	$\{q_6\}$	$\{q_6\}$
q_7	$\emptyset$	$\{q_8\}$	$\emptyset$	$\emptyset$	$\emptyset$
q_8	$\{q_9\}$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$
q_9	$\{q_{10}\}$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$
q_{10}	$\emptyset$	$\{q_{11}\}$	$\emptyset$	$\emptyset$	$\emptyset$
q_{11}	$\emptyset$	$\{q_{12}\}$	$\emptyset$	$\emptyset$	$\emptyset$
$*q_{12}$	$\{q_{12}\}$	$\{q_{12}\}$	$\{q_{12}\}$	$\{q_{12}\}$	$\{q_{12}\}$



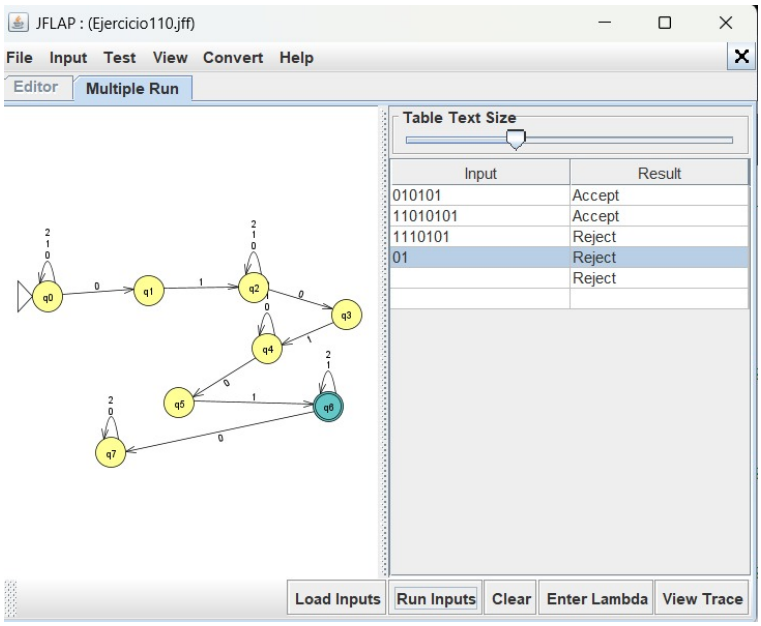
109.* Las palabras de Σ_2 que incluyen al menos una vez abbcb y al menos una vez cbaabb, permitiendo traslape.

Estado	a	b	c	0	1
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0\}$	$\{q_0, q_6\}$	$\{q_0\}$	$\{q_0\}$
q_1	$\emptyset$	$\{q_2\}$	$\emptyset$	$\emptyset$	$\emptyset$
q_2	$\emptyset$	$\{q_3\}$	$\emptyset$	$\emptyset$	$\emptyset$
q_3	$\emptyset$	$\emptyset$	$\{q_4\}$	$\emptyset$	$\emptyset$
q_4	$\emptyset$	$\{q_5\}$	$\emptyset$	$\emptyset$	$\emptyset$
q_5	$\{q_{12}\}$	$\{q_{12}\}$	$\{q_{12}\}$	$\{q_{12}\}$	$\{q_{12}\}$
q_6	$\emptyset$	$\{q_7\}$	$\emptyset$	$\emptyset$	$\emptyset$
q_7	$\{q_8\}$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$
q_8	$\{q_9\}$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$
q_9	$\emptyset$	$\{q_{10}\}$	$\emptyset$	$\emptyset$	$\emptyset$
q_{10}	$\emptyset$	$\{q_{11}\}$	$\emptyset$	$\emptyset$	$\emptyset$
q_{11}	$\{q_{12}\}$	$\{q_{12}\}$	$\{q_{12}\}$	$\{q_{12}\}$	$\{q_{12}\}$
q_{12}	$\{q_{12}, q_1\}$	$\{q_{12}\}$	$\{q_{12}, q_6\}$	$\{q_{12}\}$	$\{q_{12}\}$
$*q_{13}$	$\{q_{13}\}$	$\{q_{13}\}$	$\{q_{13}\}$	$\{q_{13}\}$	$\{q_{13}\}$



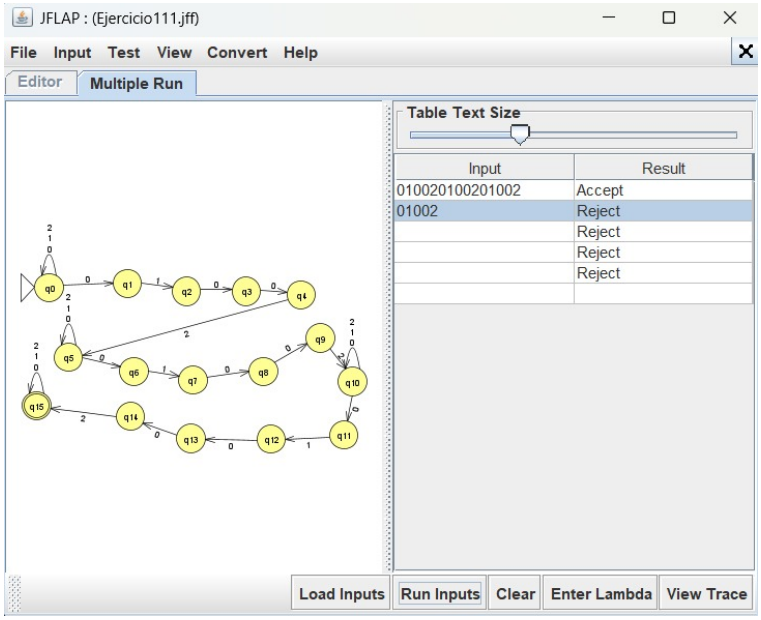
110. Las palabras de Σ_1 que contienen exactamente tres veces la subcadena 01.

Estado	0	1	2
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0\}$	$\{q_0\}$
q_1	$\emptyset$	$\{q_2\}$	$\emptyset$
q_2	$\{q_2, q_3\}$	$\{q_2\}$	$\{q_2\}$
q_3	$\emptyset$	$\{q_4\}$	$\emptyset$
q_4	$\{q_4, q_5\}$	$\{q_4\}$	$\{q_4\}$
q_5	$\emptyset$	$\{q_6\}$	$\emptyset$
$*q_6$	$\{q_7\}$	$\{q_6\}$	$\{q_6\}$
q_7	$\{q_7\}$	$\emptyset$	$\{q_6\}$



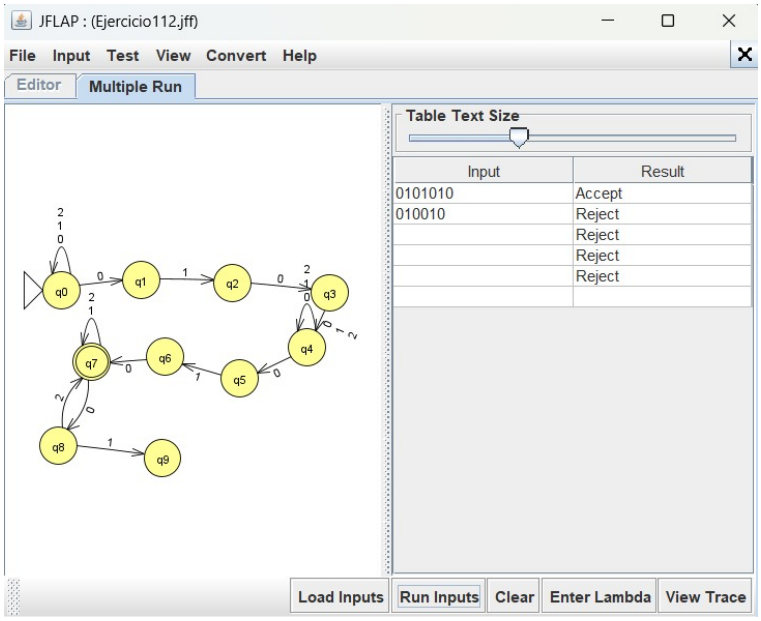
111.* Las palabras de Σ_1 que contienen exactamente tres veces la subcadena 01002.

Estado	0	1	2
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0\}$	$\{q_0\}$
q_1	$\emptyset$	$\{q_2\}$	$\emptyset$
q_2	$\{q_3\}$	$\emptyset$	$\emptyset$
q_3	$\{q_4\}$	$\emptyset$	$\emptyset$
q_4	$\emptyset$	$\emptyset$	$\{q_5\}$
q_5	$\{q_5, q_6\}$	$\{q_5\}$	$\{q_5\}$
q_6	$\emptyset$	$\{q_7\}$	$\emptyset$
q_7	$\{q_8\}$	$\emptyset$	$\emptyset$
q_8	$\{q_9\}$	$\emptyset$	$\emptyset$
q_9	$\emptyset$	$\emptyset$	$\{q_{10}\}$
q_{10}	$\{q_{10}, q_{11}\}$	$\{q_{10}\}$	$\{q_{10}\}$
q_{11}	$\emptyset$	$\{q_{12}\}$	$\emptyset$
q_{12}	$\{q_{13}\}$	$\emptyset$	$\emptyset$
q_{13}	$\{q_{14}\}$	$\emptyset$	$\emptyset$
q_{14}	$\emptyset$	$\emptyset$	$\{q_{15}\}$
$*q_{15}$	$\{q_{15}\}$	$\{q_{15}\}$	$\{q_{15}\}$



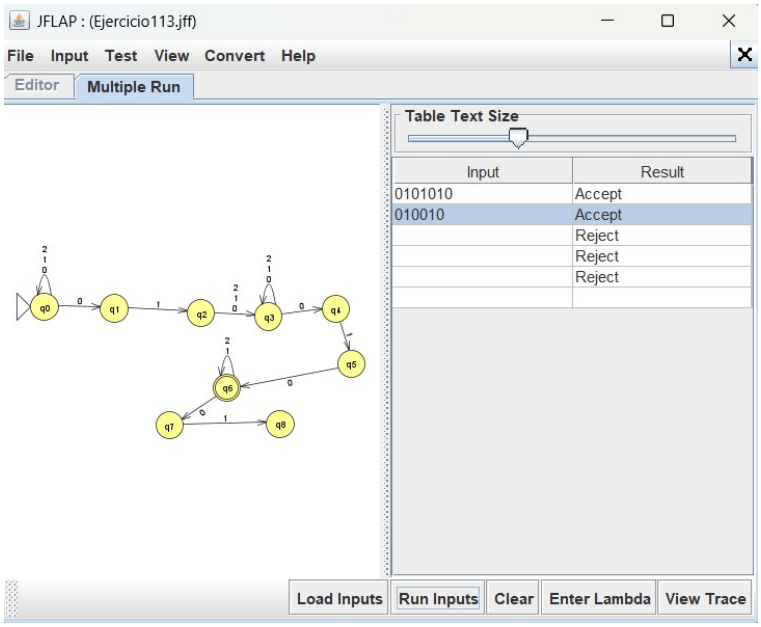
112.* Las palabras de Σ_1 que contienen dos veces la subcadena 010 sin traslape.

Estado	0	1	2
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0\}$	$\{q_0\}$
q_1	$\emptyset$	$\{q_2\}$	$\emptyset$
q_2	$\{q_3\}$	$\emptyset$	$\emptyset$
q_3	$\{q_4\}$	$\{q_4\}$	$\{q_4\}$
q_4	$\{q_4, q_5\}$	$\{q_4\}$	$\{q_4\}$
q_5	$\emptyset$	$\{q_6\}$	$\emptyset$
q_6	$\{q_7\}$	$\emptyset$	$\emptyset$
$*q_7$	$\{q_8\}$	$\{q_7\}$	$\{q_7\}$
q_8	$\emptyset$	$\{q_9\}$	$\{q_7\}$
q_9	$\emptyset$	$\emptyset$	$\emptyset$

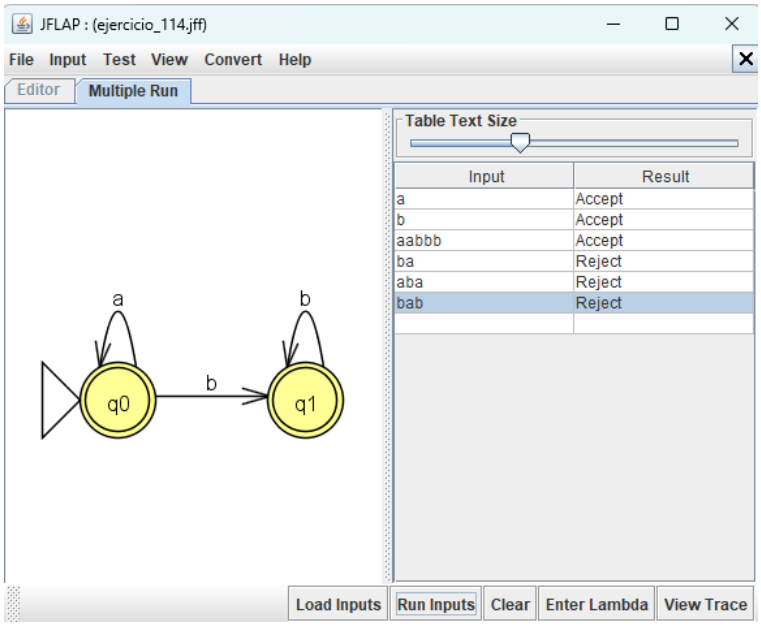


113.* Las palabras de Σ_1 que contienen dos veces la subcadena 010 permitiendo traslape.

Estado	0	1	2
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0\}$	$\{q_0\}$
q_1	$\emptyset$	$\{q_2\}$	$\emptyset$
q_2	$\{q_3\}$	$\{q_3\}$	$\{q_3\}$
q_3	$\{q_3, q_4\}$	$\{q_3\}$	$\{q_3\}$
q_4	$\emptyset$	$\{q_5\}$	$\emptyset$
q_5	$\{q_6\}$	$\emptyset$	$\emptyset$
$*q_6$	$\{q_7\}$	$\{q_6\}$	$\{q_6\}$
q_7	$\emptyset$	$\{q_8\}$	$\emptyset$
q_8	$\emptyset$	$\emptyset$	$\emptyset$

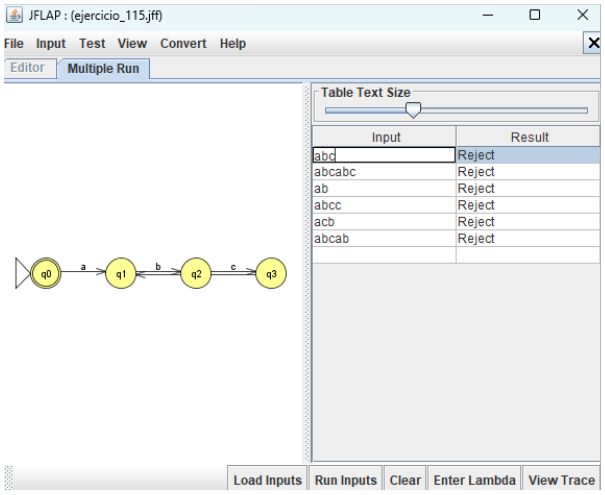

JFLAP : (Ejercicio113.jff)
File Input Test View Convert Help
Editor
Multiple Run
Table Text Size
Input
Result
0101010
Accept
010010
Accept
Reject
Reject
Reject
Load Inputs
Run Inputs
Clear
Enter Lambda
View Trace

114. $L = \{w \in \Sigma_3^* : w = a^n b^m,\ m, n \in \mathbb{N}\}$



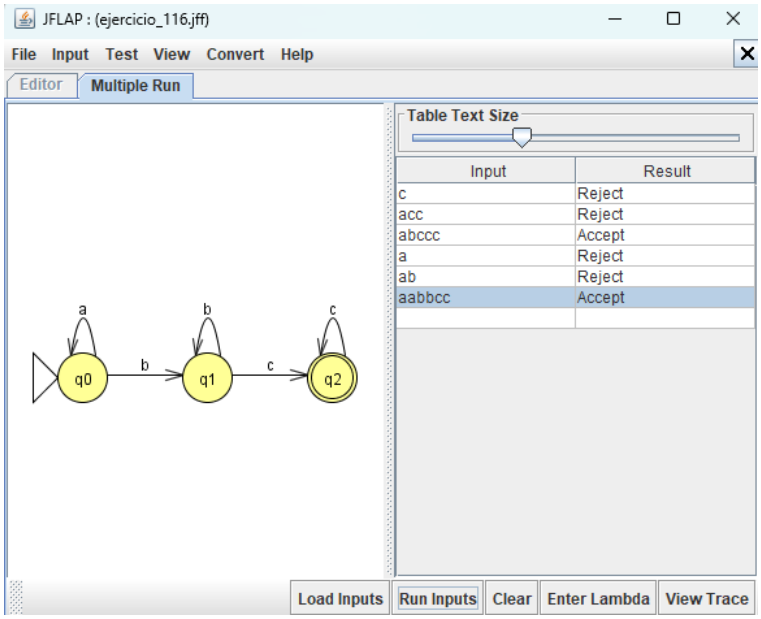
Estado	a	b
$\rightarrow *q_0$	$\{q_0\}$	$\{q_1\}$
$*q_1$	$\varnothing$	$\{q_1\}$

115. $L = \{w \in \Sigma_2^* : w = (abc)^n,\ n \in \mathbb{N}\}$



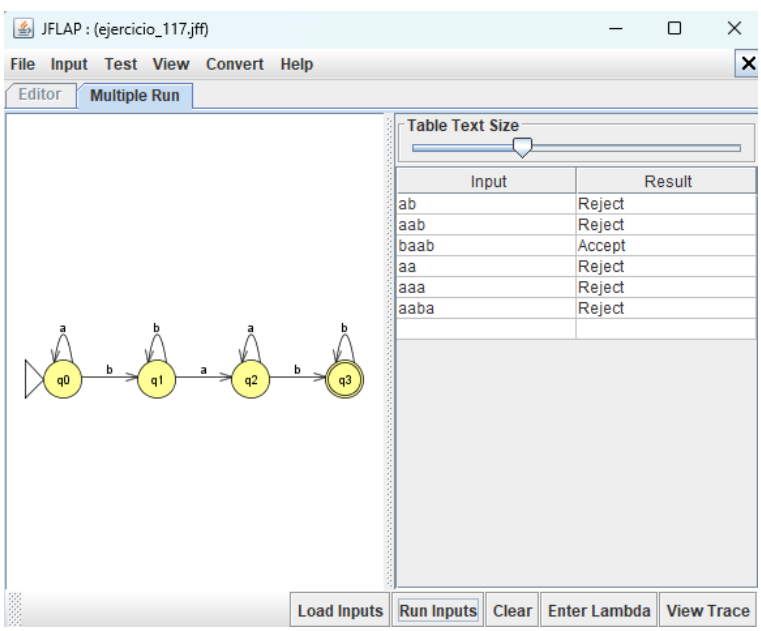
Estado	a	b	c
$\rightarrow *q_0$	$\{q_1\}$	$\varnothing$	$\varnothing$
q_1	$\varnothing$	$\{q_2\}$	$\varnothing$
q_2	$\varnothing$	$\varnothing$	$\{q_3\}$
q_3	$\{q_1\}$	$\varnothing$	$\varnothing$

116. $L = \{w \in \Sigma_2^* : w = a^n b^m c^{n+1},\ m, n \in \mathbb{N}\}$



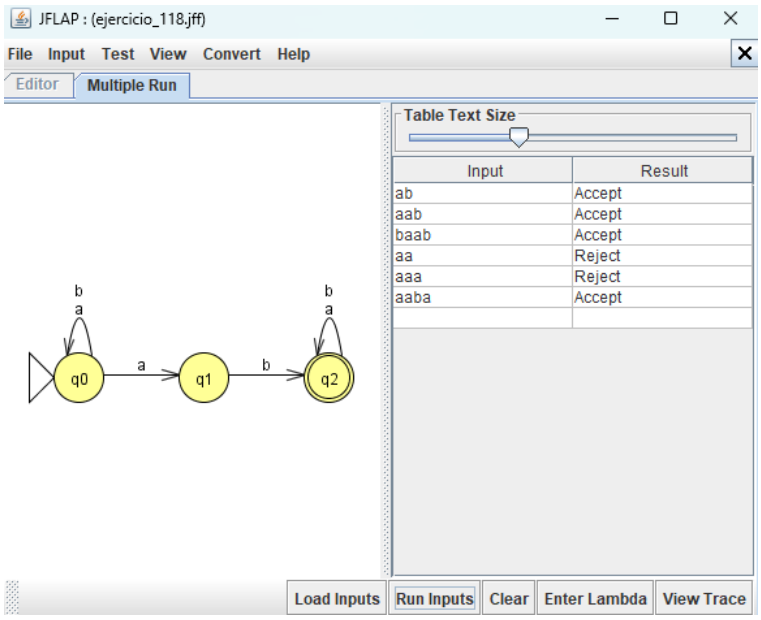
Estado	a	b	c
$\rightarrow q_0$	$\{q_0\}$	$\{q_1\}$	$\varnothing$
q_1	$\varnothing$	$\{q_1\}$	$\{q_2\}$
$*q_2$	$\varnothing$	$\varnothing$	$\{q_2\}$

117. $L = \{w \in \Sigma_3^* : w = a^n b^m a^p b^q,\ m, n, p, q \in \mathbb{N}\}$



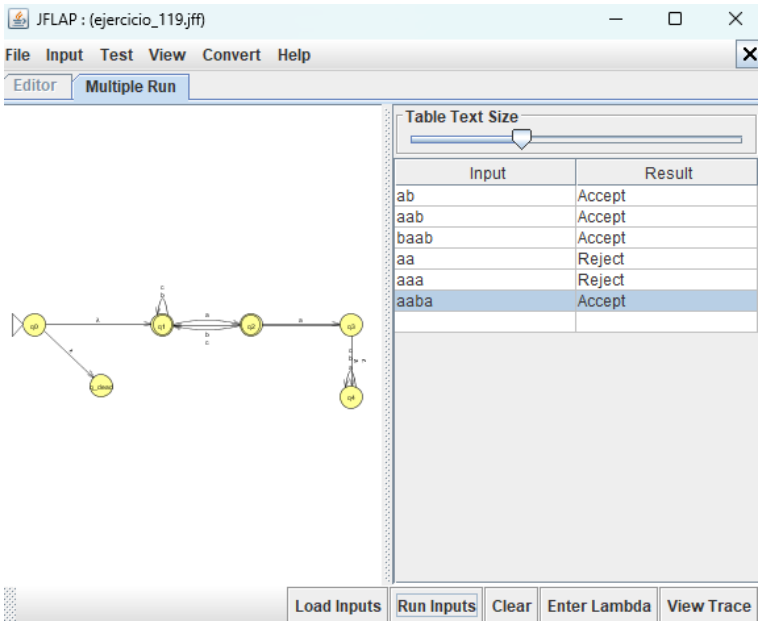
Estado	a	b
$\rightarrow q_0$	$\{q_0\}$	$\{q_1\}$
q_1	$\{q_2\}$	$\{q_1\}$
q_2	$\{q_2\}$	$\{q_3\}$
$*q_3$	$\varnothing$	$\{q_3\}$

118. $L = \{w \in \Sigma_3^* : w = axabya,\ x, y \in \Sigma_3^*\}$



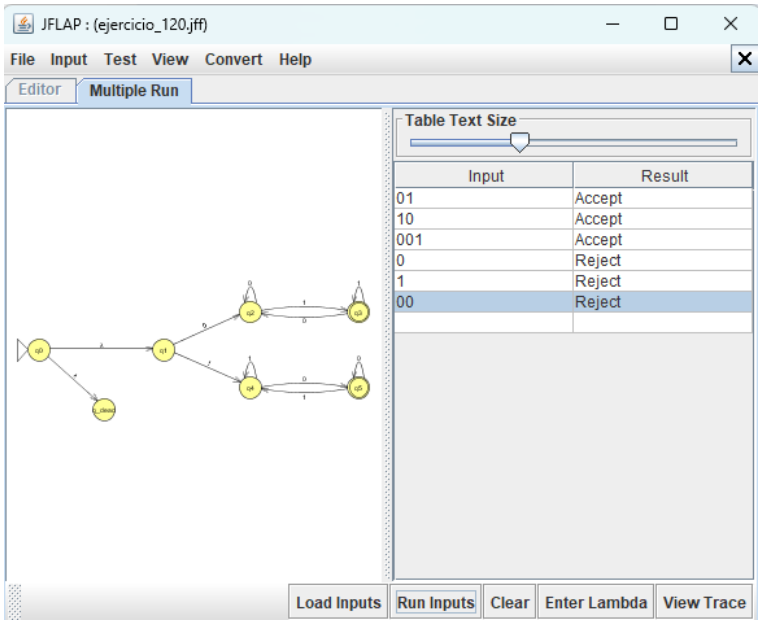
Estado	a	b
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0\}$
q_1	$\varnothing$	$\{q_2\}$
$*q_2$	$\{q_2\}$	$\{q_2\}$

119. Las palabras de Σ_3 tales que después de cada subcadena aa aparece una b



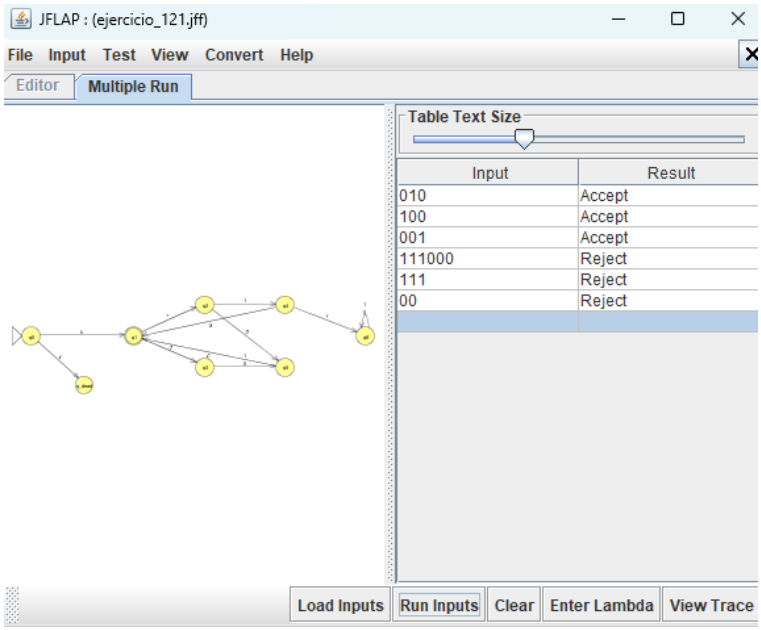
Estado	ε	a	b	c
$\rightarrow q_0$	$\{q_1, q_{dead}\}$	$\varnothing$	$\varnothing$	$\varnothing$
q_{dead}	$\varnothing$	$\varnothing$	$\varnothing$	$\varnothing$
$*q_1$	$\varnothing$	$\{q_2\}$	$\{q_1\}$	$\{q_1\}$
$*q_2$	$\varnothing$	$\{q_3\}$	$\{q_1\}$	$\{q_1\}$
q_3	$\varnothing$	$\{q_4\}$	$\{q_1\}$	$\{q_4\}$
q_4	$\varnothing$	$\{q_4\}$	$\{q_4\}$	$\{q_4\}$

120. Las palabras de Σ_1 tales que el primer carácter y el último son distintos



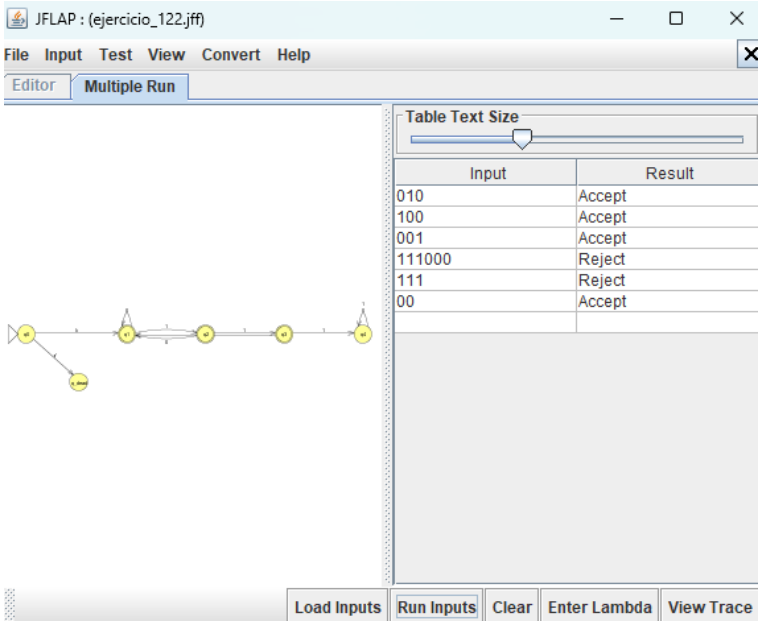
Estado	ε	0	1
$\rightarrow q_0$	$\{q_1, q_{dead}\}$	$\varnothing$	$\varnothing$
q_{dead}	$\varnothing$	$\varnothing$	$\varnothing$
q_1	$\varnothing$	$\{q_2\}$	$\{q_4\}$
q_2	$\varnothing$	$\{q_2\}$	$\{q_3\}$
$*q_3$	$\varnothing$	$\{q_2\}$	$\{q_3\}$
q_4	$\varnothing$	$\{q_5\}$	$\{q_4\}$
$*q_5$	$\varnothing$	$\{q_5\}$	$\{q_4\}$

121. Palabras sobre Σ_1 formadas por bloques de longitud 3 que contienen al menos un 0



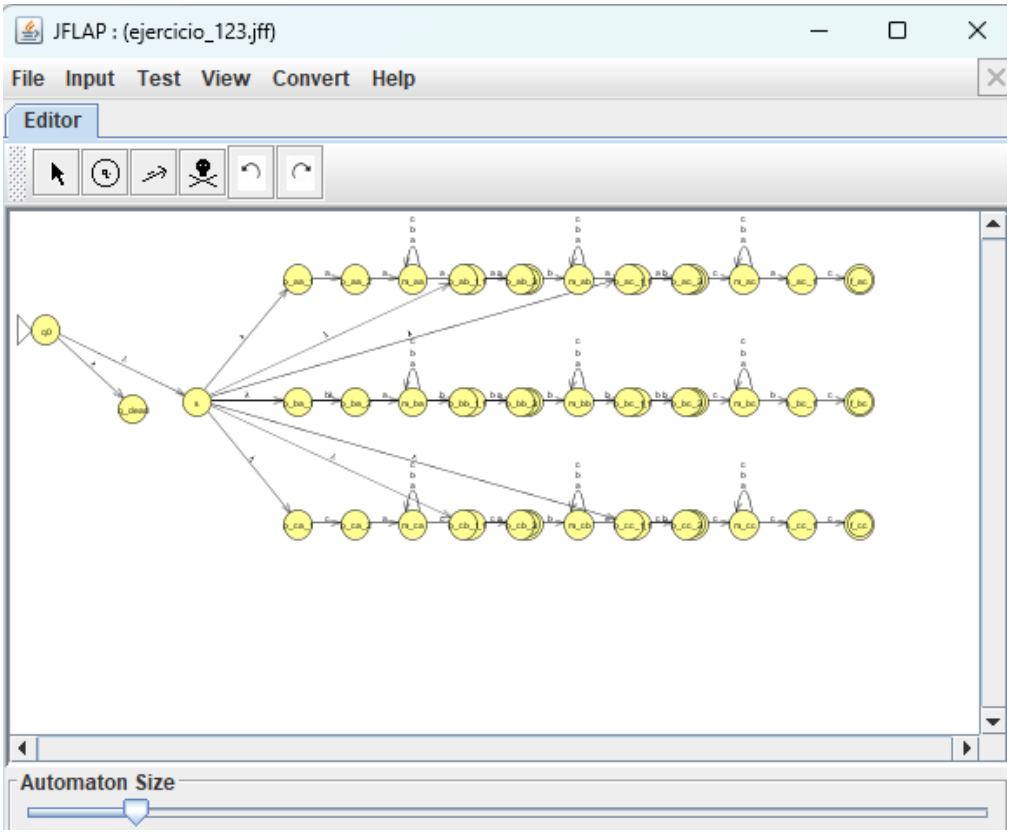
Estado	ε	0	1
$\rightarrow q_0$	$\{q_1, q_{dead}\}$	$\varnothing$	$\varnothing$
q_{dead}	$\varnothing$	$\varnothing$	$\varnothing$
$*q_1$	$\varnothing$	$\{q_3\}$	$\{q_2\}$
q_2	$\varnothing$	$\{q_5\}$	$\{q_4\}$
q_3	$\varnothing$	$\{q_5\}$	$\{q_5\}$
q_4	$\varnothing$	$\{q_1\}$	$\{q_6\}$
q_5	$\varnothing$	$\{q_1\}$	$\{q_1\}$
q_6	$\varnothing$	$\{q_6\}$	$\{q_6\}$

122. Toda subcadena de longitud 3 contiene al menos un 0



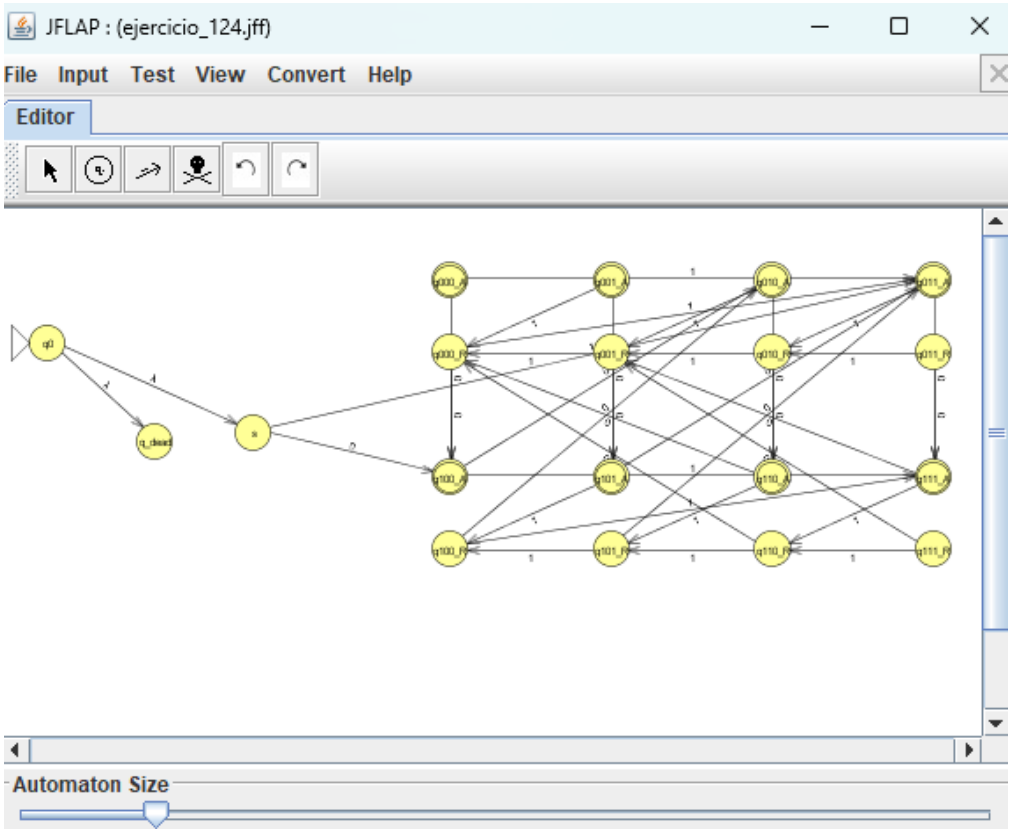
Estado	ε	0	1
$\rightarrow q_0$	$\{q_1, q_{dead}\}$	$\varnothing$	$\varnothing$
q_{dead}	$\varnothing$	$\varnothing$	$\varnothing$
$*q_1$	$\varnothing$	$\{q_1\}$	$\{q_2\}$
$*q_2$	$\varnothing$	$\{q_1\}$	$\{q_3\}$
$*q_3$	$\varnothing$	$\{q_1\}$	$\{q_4\}$
q_4	$\varnothing$	$\{q_4\}$	$\{q_4\}$

123. $L = \{wxw : w \in \Sigma_3^2,\ x \in \Sigma_3^*\}$



Estado	ε	a	b	c
$\rightarrow q_0$	$\{s, q_{dead}\}$	$\varnothing$	$\varnothing$	$\varnothing$
q_{dead}	$\varnothing$	$\varnothing$	$\varnothing$	$\varnothing$
s	$\{p_{aa1}, p_{ab1}, p_{ac1}, p_{ba1}, p_{bb1}, p_{bc1}, p_{ca1}, p_{cb1}, p_{cc1}\}$	$\varnothing$	$\varnothing$	$\varnothing$
p_{aa1}	$\varnothing$	$\{p_{aa2}\}$	$\varnothing$	$\varnothing$
p_{aa2}	$\varnothing$	$\{m_{aa}\}$	$\varnothing$	$\varnothing$
m_{aa}	$\varnothing$	$\{m_{aa}, s_{aa1}\}$	$\{m_{aa}\}$	$\{m_{aa}\}$
s_{aa1}	$\varnothing$	$\{f_{aa}\}$	$\varnothing$	$\varnothing$
$*f_{aa}$	$\varnothing$	$\varnothing$	$\varnothing$	$\varnothing$
p_{ab1}	$\varnothing$	$\{p_{ab2}\}$	$\varnothing$	$\varnothing$
p_{ab2}	$\varnothing$	$\varnothing$	$\{m_{ab}\}$	$\varnothing$
m_{ab}	$\varnothing$	$\{m_{ab}, s_{ab1}\}$	$\{m_{ab}\}$	$\{m_{ab}\}$
s_{ab1}	$\varnothing$	$\varnothing$	$\{f_{ab}\}$	$\varnothing$
$*f_{ab}$	$\varnothing$	$\varnothing$	$\varnothing$	$\varnothing$
p_{ac1}	$\varnothing$	$\{p_{ac2}\}$	$\varnothing$	$\varnothing$
p_{ac2}	$\varnothing$	$\varnothing$	$\varnothing$	$\{m_{ac}\}$
m_{ac}	$\varnothing$	$\{m_{ac}, s_{ac1}\}$	$\{m_{ac}\}$	$\{m_{ac}\}$
s_{ac1}	$\varnothing$	$\varnothing$	$\varnothing$	$\{f_{ac}\}$
$*f_{ac}$	$\varnothing$	$\varnothing$	$\varnothing$	$\varnothing$
p_{ba1}	$\varnothing$	$\varnothing$	$\{p_{ba2}\}$	$\varnothing$
p_{ba2}	$\varnothing$	$\{m_{ba}\}$	$\varnothing$	$\varnothing$
m_{ba}	$\varnothing$	$\{m_{ba}\}$	$\{m_{ba}, s_{ba1}\}$	$\{m_{ba}\}$
s_{ba1}	$\varnothing$	$\{f_{ba}\}$	$\varnothing$	$\varnothing$
$*f_{ba}$	$\varnothing$	$\varnothing$	$\varnothing$	$\varnothing$
p_{bb1}	$\varnothing$	$\varnothing$	$\{p_{bb2}\}$	$\varnothing$
p_{bb2}	$\varnothing$	$\varnothing$	$\{m_{bb}\}$	$\varnothing$
m_{bb}	$\varnothing$	$\{m_{bb}\}$	$\{m_{bb}, s_{bb1}\}$	$\{m_{bb}\}$
s_{bb1}	$\varnothing$	$\varnothing$	$\{f_{bb}\}$	$\varnothing$
$*f_{bb}$	$\varnothing$	$\varnothing$	$\varnothing$	$\varnothing$
p_{bc1}	$\varnothing$	$\varnothing$	$\{p_{bc2}\}$	$\varnothing$
p_{bc2}	$\varnothing$	$\varnothing$	$\varnothing$	$\{m_{bc}\}$
m_{bc}	$\varnothing$	$\{m_{bc}\}$	$\{m_{bc}, s_{bc1}\}$	$\{m_{bc}\}$
s_{bc1}	$\varnothing$	$\varnothing$	$\varnothing$	$\{f_{bc}\}$
$*f_{bc}$	$\varnothing$	$\varnothing$	$\varnothing$	$\varnothing$
p_{ca1}	$\varnothing$	$\varnothing$	$\varnothing$	$\{p_{ca2}\}$
p_{ca2}	$\varnothing$	$\{m_{ca}\}$	$\varnothing$	$\varnothing$
m_{ca}	$\varnothing$	$\{m_{ca}\}$	$\{m_{ca}\}$	$\{m_{ca}, s_{ca1}\}$
s_{ca1}	$\varnothing$	$\{f_{ca}\}$	$\varnothing$	$\varnothing$
$*f_{ca}$	$\varnothing$	$\varnothing$	$\varnothing$	$\varnothing$
p_{cb1}	$\varnothing$	$\varnothing$	$\varnothing$	$\{p_{cb2}\}$
p_{cb2}	$\varnothing$	$\varnothing$	$\{m_{cb}\}$	$\varnothing$
m_{cb}	$\varnothing$	$\{m_{cb}\}$	$\{m_{cb}\}$	$\{m_{cb}, s_{cb1}\}$
s_{cb1}	$\varnothing$	$\varnothing$	$\{f_{cb}\}$	$\varnothing$
$*f_{cb}$	$\varnothing$	$\varnothing$	$\varnothing$	$\varnothing$
p_{cc1}	$\varnothing$	$\varnothing$	$\varnothing$	$\{p_{cc2}\}$
p_{cc2}	$\varnothing$	$\varnothing$	$\varnothing$	$\{m_{cc}\}$
m_{cc}	$\varnothing$	$\{m_{cc}\}$	$\{m_{cc}\}$	$\{m_{cc}, s_{cc1}\}$
s_{cc1}	$\varnothing$	$\varnothing$	$\varnothing$	$\{f_{cc}\}$
$*f_{cc}$	$\varnothing$	$\varnothing$	$\varnothing$	$\varnothing$

124. Sistema de palancas



Estado	ε	0	1
$\rightarrow q_0$	$\{s, q_{dead}\}$	$\varnothing$	$\varnothing$
q_{dead}	$\varnothing$	$\varnothing$	$\varnothing$
s	$\varnothing$	$\{q100_A\}$	$\{q011_A\}$
$*q000_A$	$\varnothing$	$\{q100_A\}$	$\{q011_A\}$
$q000_R$	$\varnothing$	$\{q100_A\}$	$\{q011_A\}$
$*q001_A$	$\varnothing$	$\{q101_A\}$	$\{q000_R\}$
$q001_R$	$\varnothing$	$\{q101_A\}$	$\{q000_R\}$
$*q010_A$	$\varnothing$	$\{q110_A\}$	$\{q001_R\}$
$q010_R$	$\varnothing$	$\{q110_A\}$	$\{q001_R\}$
$*q011_A$	$\varnothing$	$\{q111_A\}$	$\{q010_R\}$
$q011_R$	$\varnothing$	$\{q111_A\}$	$\{q010_R\}$
$*q100_A$	$\varnothing$	$\{q010_A\}$	$\{q111_A\}$
$q100_R$	$\varnothing$	$\{q010_A\}$	$\{q111_A\}$
$*q101_A$	$\varnothing$	$\{q011_A\}$	$\{q100_R\}$
$q101_R$	$\varnothing$	$\{q011_A\}$	$\{q100_R\}$
$*q110_A$	$\varnothing$	$\{q000_R\}$	$\{q101_R\}$
$q110_R$	$\varnothing$	$\{q000_R\}$	$\{q101_R\}$
$*q111_A$	$\varnothing$	$\{q001_R\}$	$\{q110_R\}$
$q111_R$	$\varnothing$	$\{q001_R\}$	$\{q110_R\}$

125.

Si $F = Q$, entonces el lenguaje aceptado es Σ^*.

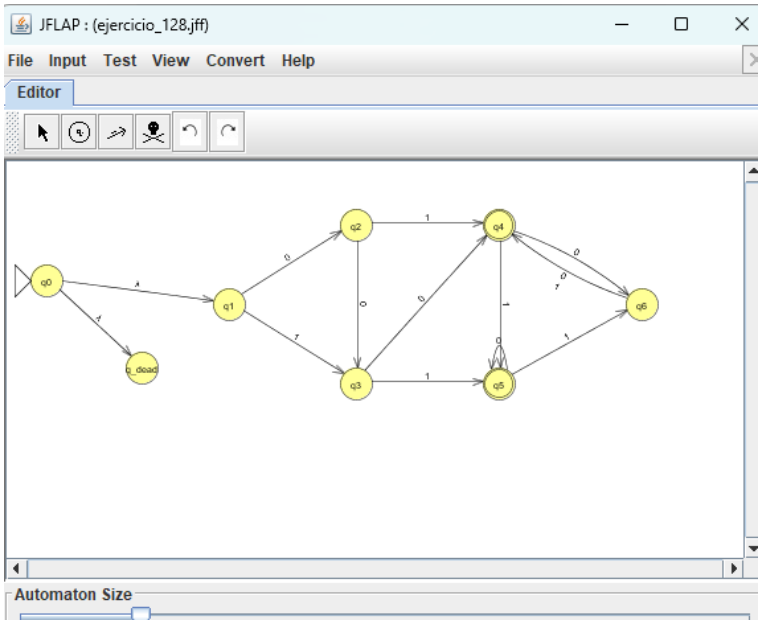
126.

Si $F = \varnothing$, entonces el lenguaje aceptado es $\varnothing$.

127.

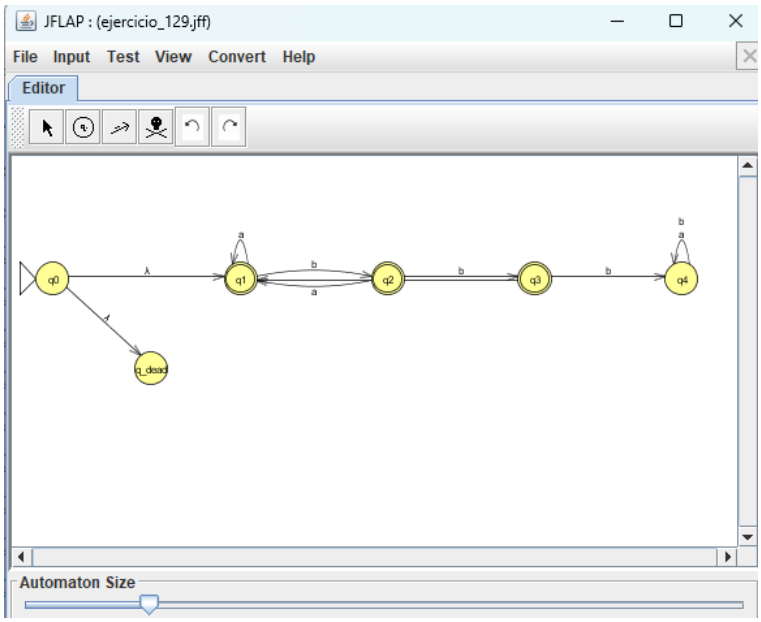
Si $F = Q$, entonces el lenguaje aceptado es Σ^*.

128. Quitar el último símbolo



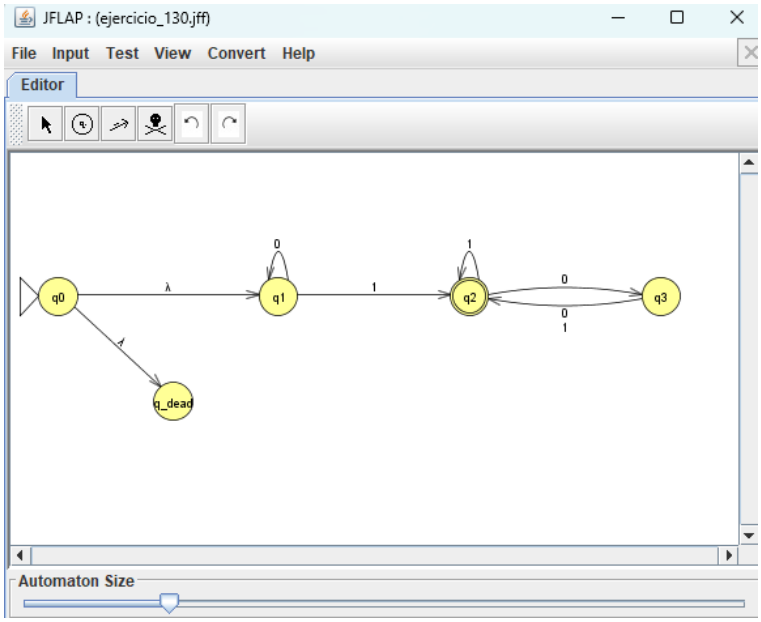
Estado	ε	0	1
$\rightarrow q_0$	$\{q_1, q_{dead}\}$	$\varnothing$	$\varnothing$
q_{dead}	$\varnothing$	$\varnothing$	$\varnothing$
q_1	$\varnothing$	$\{q_2\}$	$\{q_3\}$
q_2	$\varnothing$	$\{q_3\}$	$\{q_4\}$
q_3	$\varnothing$	$\{q_4\}$	$\{q_5\}$
$*q_4$	$\varnothing$	$\{q_6\}$	$\{q_5\}$
$*q_5$	$\varnothing$	$\{q_5\}$	$\{q_6\}$
q_6	$\varnothing$	$\{q_4\}$	$\{q_4\}$

129. No aparecen tres b consecutivas



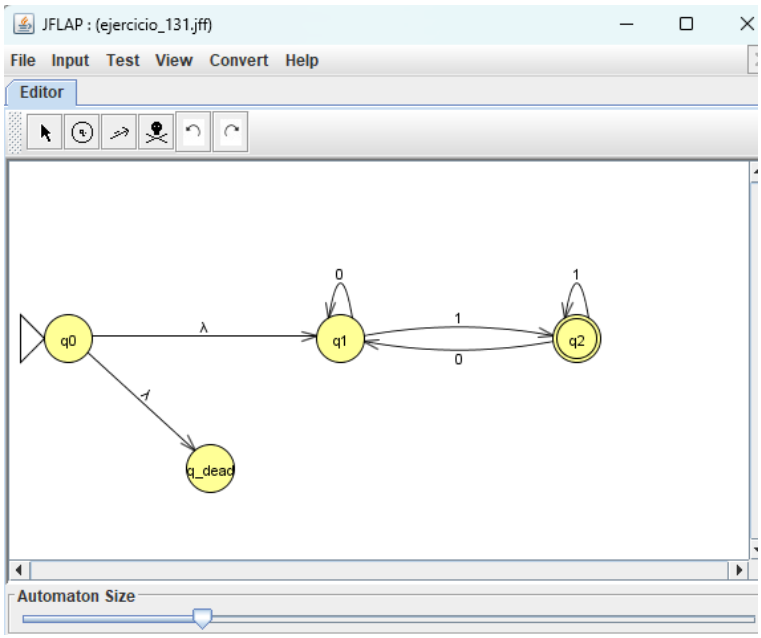
Estado	ε	a	b
$\rightarrow q_0$	$\{q_1, q_{dead}\}$	$\varnothing$	$\varnothing$
q_{dead}	$\varnothing$	$\varnothing$	$\varnothing$
$*q_1$	$\varnothing$	$\{q_1\}$	$\{q_2\}$
$*q_2$	$\varnothing$	$\{q_1\}$	$\{q_3\}$
$*q_3$	$\varnothing$	$\{q_1\}$	$\{q_4\}$
q_4	$\varnothing$	$\{q_4\}$	$\{q_4\}$

130.



Estado	ε	0	1
$\rightarrow q_0$	$\{q_1, q_{dead}\}$	$\varnothing$	$\varnothing$
q_{dead}	$\varnothing$	$\varnothing$	$\varnothing$
q_1	$\varnothing$	$\{q_1\}$	$\{q_2\}$
$*q_2$	$\varnothing$	$\{q_3\}$	$\{q_2\}$
q_3	$\varnothing$	$\{q_2\}$	$\{q_2\}$

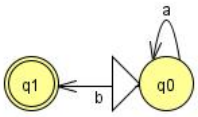
131.



Estado	ε	0	1
$\rightarrow q_0$	$\{q_1, q_{dead}\}$	$\varnothing$	$\varnothing$
q_{dead}	$\varnothing$	$\varnothing$	$\varnothing$
q_1	$\varnothing$	$\{q_1\}$	$\{q_2\}$
$*q_2$	$\varnothing$	$\{q_1\}$	$\{q_2\}$

132. $L = \{a^n b : n \geq 0\}$

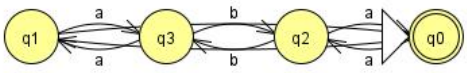
	a	b
$\rightarrow q_0$	$\{q_0\}$	$\{q_1\}$
$*q_1$	$\emptyset$	$\emptyset$



133. Tabla de estados y diagrama de transiciones
Procedimiento:
Se construye la tabla con base en la función de transición dada:

δ	a	b
q_0	$\{q_2\}$	$\{q_1\}$
q_1	$\{q_3\}$	$\{q_0\}$
q_2	$\{q_0\}$	$\{q_3\}$
q_3	$\{q_1\}$	$\{q_2\}$

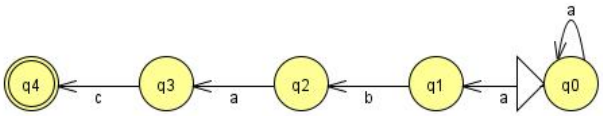
Estado inicial: q_0
Estado de aceptación: q_0



134. Lenguaje que contiene el patrón *abac*
Procedimiento:
Se construye un AFD que detecta la subcadena *abac*. Cada estado representa el avance en el reconocimiento del patrón. Una vez detectado, el autómata permanece en aceptación.

134. Contiene .ªbac"

	a	b	c
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0\}$	$\{q_0\}$
q_1	$\emptyset$	$\{q_2\}$	$\emptyset$
q_2	$\{q_3\}$	$\emptyset$	$\emptyset$
q_3	$\emptyset$	$\emptyset$	$\{q_4\}$
$*q_4$	$\{q_4\}$	$\{q_4\}$	$\{q_4\}$



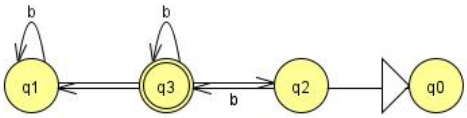
135. $L = \{b^m a b^n : m, n > 0\}$

Procedimiento:

El autómata exige al menos una b, seguida de una a, y luego al menos una b. Se utilizan estados para verificar cada parte de la cadena.

135. $L = \{b^m a b^n : m, n > 0\}$

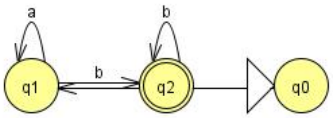
	a	b
$\rightarrow q_0$	$\emptyset$	$\{q_1\}$
q_1	$\{q_2\}$	$\{q_1\}$
q_2	$\emptyset$	$\{q_3\}$
$*q_3$	$\emptyset$	$\{q_3\}$



136. $L = \{a^m b^n : m, n > 0\}$
Procedimiento:
Se requiere al menos una a seguida de al menos una b. No se permite regresar a a después de haber leído b.

136. $L = \{a^m b^n : m, n > 0\}$

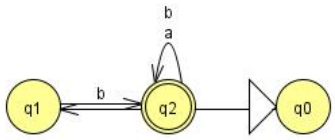
	a	b
$\rightarrow q_0$	$\{q_1\}$	$\emptyset$
q_1	$\{q_1\}$	$\{q_2\}$
$*q_2$	$\emptyset$	$\{q_2\}$



137. Cadenas que comienzan con "ab"
Procedimiento:
El autómata verifica que los dos primeros símbolos sean a y b. Después de eso, acepta cualquier combinación de símbolos.

137. Prefijo .ªb"

	a	b
$\rightarrow q_0$	$\{q_1\}$	$\emptyset$
q_1	$\emptyset$	$\{q_2\}$
$*q_2$	$\{q_2\}$	$\{q_2\}$



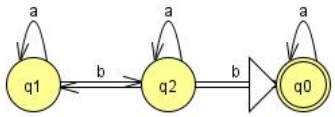
138. Número de b divisible entre 3

Procedimiento:

Se utilizan tres estados que representan residuos módulo 3. El autómata cambia de estado solo al leer b.

	a	b
$*q_0$	$\{q_0\}$	$\{q_1\}$
q_1	$\{q_1\}$	$\{q_2\}$
q_2	$\{q_2\}$	$\{q_0\}$

q_0 es el estado de aceptación.



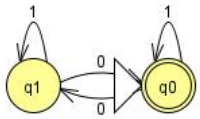
139. Número par de ceros

Procedimiento:

Se usan dos estados: uno para cantidad par y otro para impar. Cada vez que se lee un 0, se cambia de estado.

139. Número par de 0's

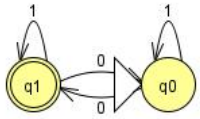
	0	1
$\rightarrow *q_0$	$\{q_1\}$	$\{q_0\}$
q_1	$\{q_0\}$	$\{q_1\}$



140.
a) Número impar de ceros
Se utiliza un AFD de dos estados alternando con cada 0.
b) Número par de ceros y unos
Se construye un AFD con cuatro estados combinando paridad de ambos.

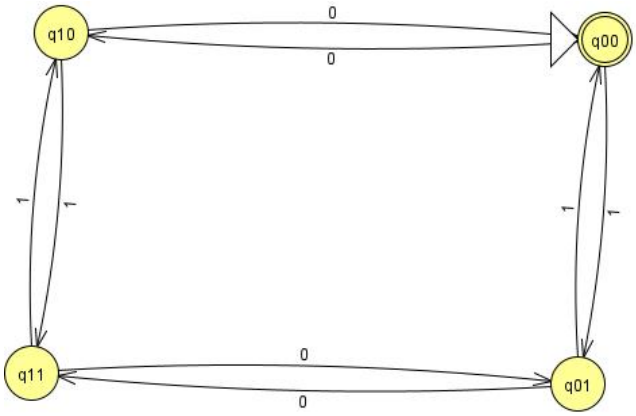
140.
a) Impar de 0's

	0	1
$\rightarrow q_0$	$\{q_1\}$	$\{q_0\}$
$*q_1$	$\{q_0\}$	$\{q_1\}$



b) Par de 0's y 1's

	0	1
$\rightarrow *q_{00}$	$\{q_{10}\}$	$\{q_{01}\}$
q_{10}	$\{q_{00}\}$	$\{q_{11}\}$
q_{01}	$\{q_{11}\}$	$\{q_{00}\}$
q_{11}	$\{q_{01}\}$	$\{q_{10}\}$



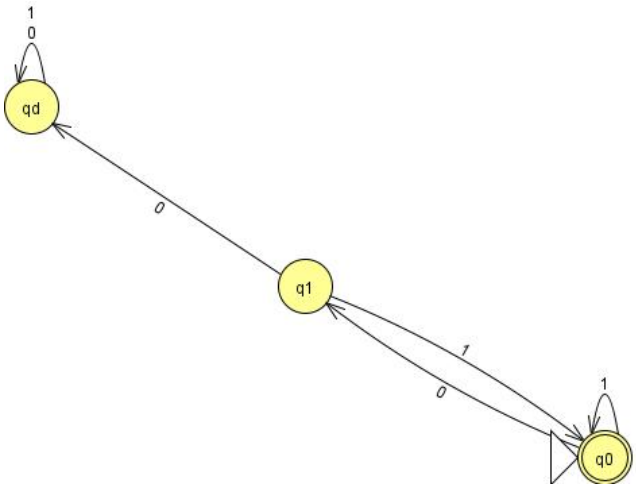
141. Cada 0 tiene un 1 a la derecha

Procedimiento:

Se asegura que después de cada 0 aparezca inmediatamente un 1. Si no ocurre, se pasa a un estado trampa.

141. Cada 0 seguido de 1

	0	1
$\rightarrow q_0$	$\{q_1\}$	$\{q_0\}$
q_1	$\emptyset$	$\{q_0\}$
$*q_0$	$\{q_1\}$	$\{q_0\}$



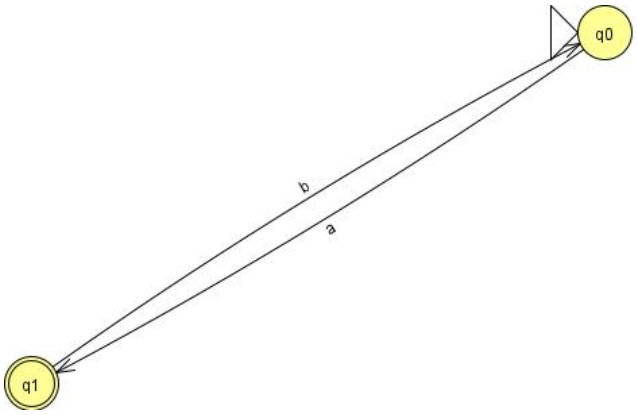
142.

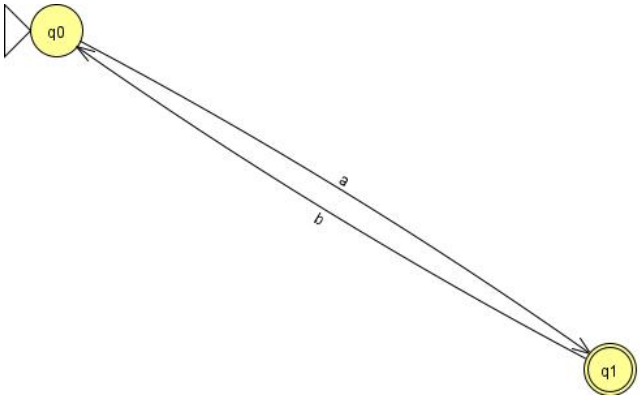
a) $(ab)^*$ **sin** ϵ

142.

(ab)

	a	b
$\rightarrow q_0$	$\{q_1\}$	$\emptyset$
q_1	$\emptyset$	$\{q_2\}$
$*q_2$	$\{q_1\}$	$\emptyset$





143.

Procedimiento:

Se analizan los autómatas dados en las figuras para identificar el lenguaje que generan, observando estados de aceptación y transiciones.

144.

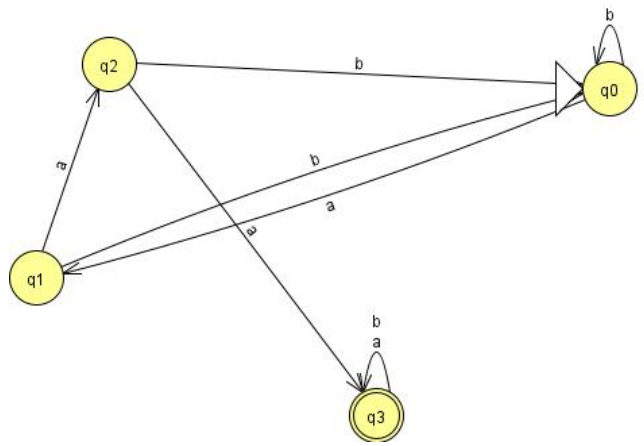
a) Cadenas que comienzan con a

b) Cadenas que comienzan con a y terminan con b

c) Cadenas que contienen aaa

d) Números enteros

e) Números enteros con signo



145. Binarios múltiplos de 5 (derecha a izquierda)

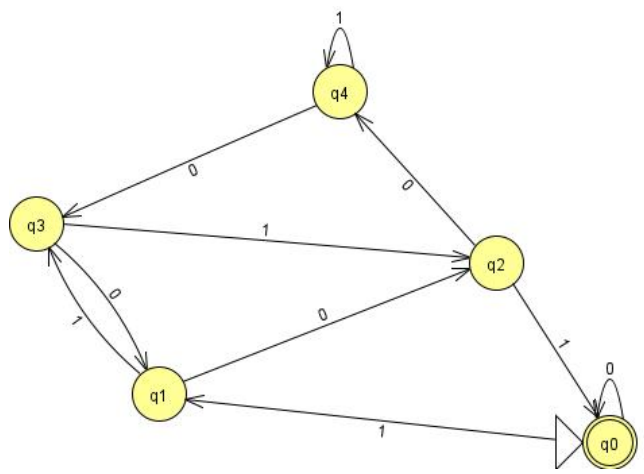
Procedimiento:

Se construye un AFN basado en residuos módulo 5, leyendo desde el bit menos significativo.

145. Múltiplos de 5 (residuos)

$$q_i \xrightarrow{0} \{q_{2i \text{ mód } 5}\}, \quad q_i \xrightarrow{1} \{q_{(2i+1) \text{ mód } 5}\}$$

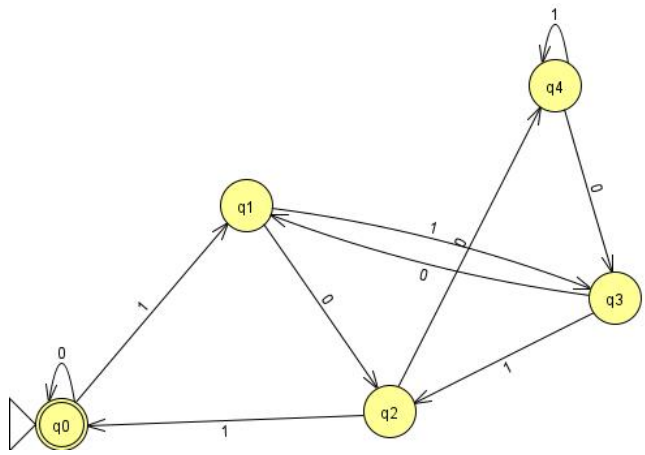
Estado de aceptación: q_0



146. Binarios múltiplos de 5 (izquierda a derecha)

Procedimiento:

Se utiliza un AFN que calcula el residuo módulo 5 conforme se leen los bits. Es el mismo que el 145 solo que deberia ser izq-der

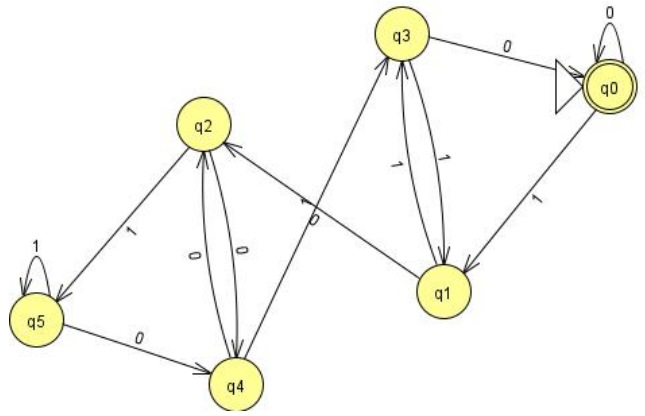


147. Binarios múltiplos de 6
Procedimiento:
Se combinan condiciones de divisibilidad por 2 y 3 mediante estados.

147. Múltiplos de 6

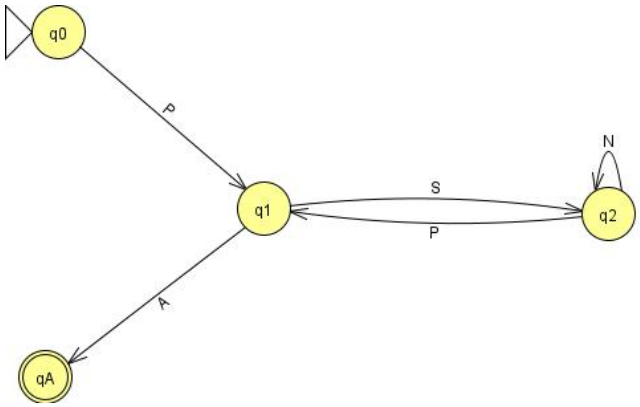
$$q_i \xrightarrow{0} \{q_{2i \text{ mód } 6}\}, \quad q_i \xrightarrow{1} \{q_{(2i+1) \text{ mód } 6}\}$$

Estado de aceptación: q_0

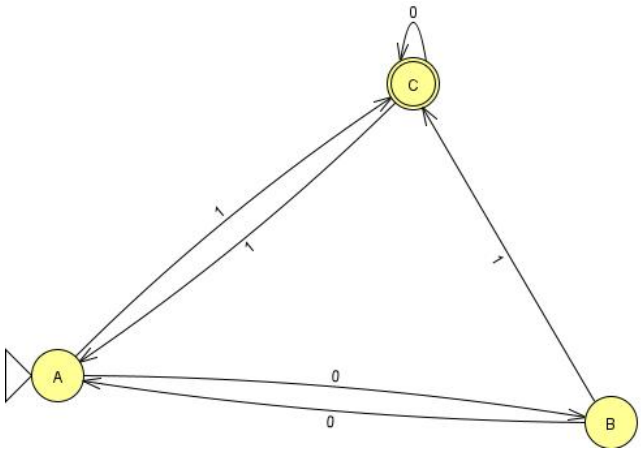


148. Modelo de alumno ESCOM
Procedimiento:
Se modelan estados que representan: - Presentación - Aprobación - Reprobación
Se controla que no haya más de dos intentos antes de reiniciar.

	P	N	A	S
$\rightarrow q_0$	$\{q_1\}$	$\{q_0\}$	$\emptyset$	$\emptyset$
q_1	$\emptyset$	$\emptyset$	$\{q_A\}$	$\{q_2\}$
q_2	$\{q_1\}$	$\{q_2\}$	$\emptyset$	$\emptyset$
$*q_A$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$

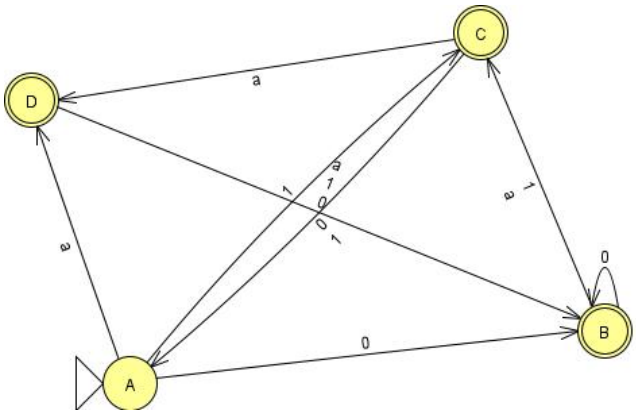


149.
Resultados:
a) $\delta(B, 0) =$ A b) $\delta(C, 1) =$ A c) $\hat{\delta}(A, 1101) =$ resultado según recorrido d) $\hat{\delta}(A, 01001)$ = resultado según recorrido



150.
Resultados:
a) $\hat{\delta}(B, 10a11)$ b) $\hat{\delta}(B, aa1100)$ c) $\hat{\delta}(A, a01a01)$ d) $\hat{\delta}(C, a11a00)$
Los valores se obtienen aplicando la función de transición extendida.

$$\delta(q, a) = \{resultado\}$$


C
D
B
A
a
a
a
a
1
1
0
0
1
0
0